

$\therefore$ We shall prove $(p \wedge \neg q) \rightarrow F$ is a True proposition.

Let, $(p \wedge \neg q)$ be True. By defn. of AND, p is True and $\neg q$ is True.

$\neg q: (n+2)$ is a perfect square.

$\therefore n$ is also a perfect square and $(n+2)$ is also a perfect square.

By defn. of perfect square, there exists integers a and b such that

$$a^2 = n \text{ and } b^2 = (n+2).$$

$\therefore b^2 - a^2 = 2 \Rightarrow (b-a)(b+a) = 2$. It's easy to see that $b-a$ and $b+a$ are both integers. 2 has only two factors 1 and 2.

Case 1: $b-a=1$ $b+a=2$ $\therefore 2b=3 \Rightarrow b=1.5$ which is not an integer

Case 2: $b+a=1$ $b-a=2$ $\therefore 2b=3 \Rightarrow b=1.5$ which is not an integer

In both cases, we arrive at the conclusion that b is not an integer. $\therefore b^2$ is not an integer, and $(n+2)$ is not a perfect square.

$\therefore \neg q$ is False. $(p \wedge \neg q)$ is False by defn. of AND. $(p \wedge \neg q) \rightarrow F$ is True by defn. of conditional.

$\therefore p \rightarrow \neg q$ is True

Since $n+2$ is not a perfect square, q is True $\square$

9) Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.

Soln: $p: x$ is an arbitrary irrational no. $q: y$ is an arbitrary rational no.
 $r: x+y$ is an irrational no.

We shall prove $(p \wedge q) \rightarrow r$ is True by contradiction

$\therefore$ We shall prove $((p \wedge q) \wedge \neg r) \rightarrow F$ is a True proposition.

Let, $(p \wedge q) \wedge \neg r$ be True. $\therefore (p \wedge q)$ is True and $\neg r$ is True (Defn. of AND)

$\neg r: x+y$ is not an irrational no, or $x+y$ is a rational no. (By defn. 2)

By defn. of AND, p is True and q is True.